



Ok I need to see this to my mate that is doing the programming so he know exactly what the setup is. I am using this to do the conversion from Raspberry pi to hub75: Conversion board for for Raspberry Pi to HUB75 for Raspberry Pi GPIO controlled RGB LED display

- AliExpress 502. I have the panels wired as follows:

Panel 1 is the upper left hand corner, which is wired to panel two which is in the uper righthand corner - the data for these panel when looking at the screen from the front is going left to right, then panel 2 is connected to the panel in the bottom right hand corner and this panel is connect to the panel in the bottom left hand corner so the data is not going right to left. Please detail this and how it could be setup/programmed in the Raspberry Pi.

Perfect! Here's a complete setup document for your mate to show exactly what you have wired and how to program it:

LED Matrix Setup Documentation

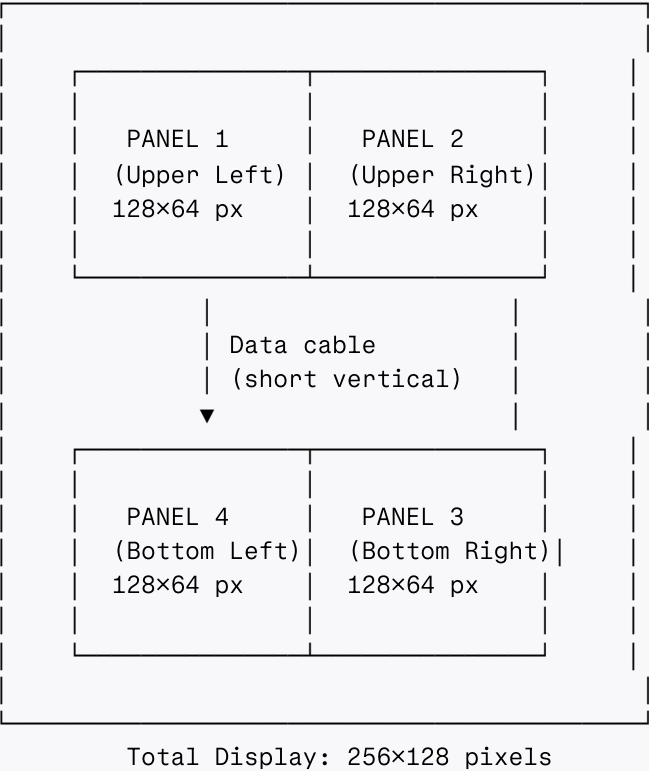
Hardware Configuration

Components:

- 4 P2 LED panels (128×64 pixels each, 256×128mm physical size)^[1]
- Raspberry Pi 4 (GPIO header connected to HUB75 adapter)
- HUB75 Adapter Board: AliExpress "Conversion board for Raspberry Pi to HUB75" [AliExpress link]
- External 5V power supply (panels powered separately, not from Pi)

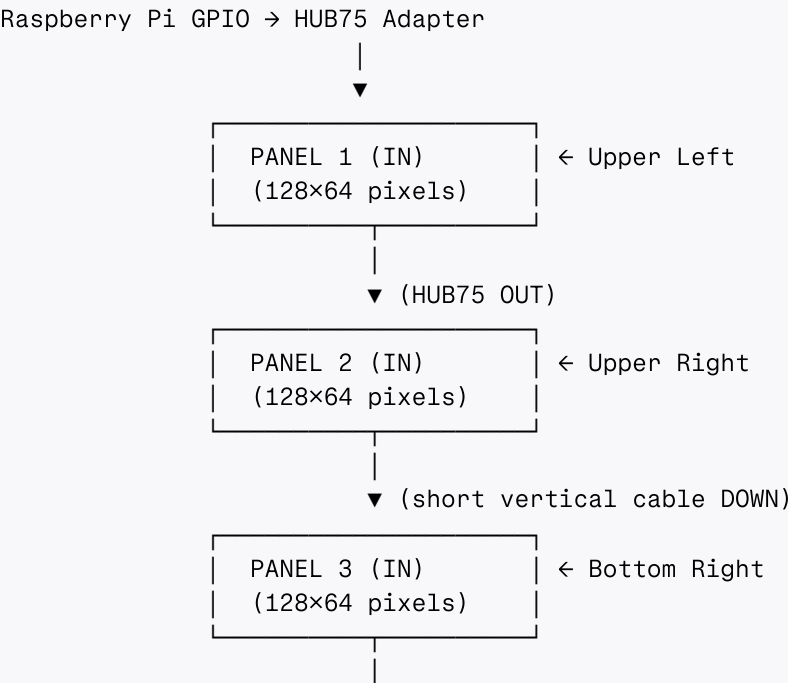
Physical Panel Layout (2×2 Grid)

FRONT VIEW (what people see when looking at display):

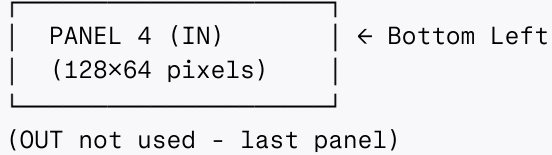


Data Flow (Critical for Programming)

Panel Wiring Chain:



▼ (HUB75 OUT, going LEFT)



Data Direction From Front View:

Top Row: PANEL 1 → PANEL 2
(left → right) ✓ Normal

Bottom Row: PANEL 3 → PANEL 4
(right → left) ✓ SERPENTINE (reversed)

Key point: The bottom row data flows **right-to-left** instead of left-to-right. This is called **SERPENTINE wiring**.

Required Software Configuration

Library: `rpi-rgb-led-matrix`

Install if not already installed:

```
cd ~
sudo apt-get update
sudo apt-get install -y git build-essential
git clone https://github.com/hzeller/rpi-rgb-led-matrix.git
cd rpi-rgb-led-matrix
sudo make install
```

Testing Command:

```
cd ~/rpi-rgb-led-matrix/utils

# Test with a sample image
sudo ./led-image-viewer \
  --led-rows=64 \
  --led-cols=128 \
  --led-chain=4 \
  --led-parallel=1 \
  --led-pixel-mapper="U-mapper" \
  --led-slowdown-gpio=4 \
  --led-row-addr-type=3 \
  --led-multiplexing=0 \
  --led-pwm-bits=7 \
  --led-brightness=70 \
```

```
--led-sba=1 \  
/path/to/test-image.png
```

Key Parameters Explained:

Parameter	Value	Why
--led-rows=64	64	Height of one panel (not total)
--led-cols=128	128	Width of one panel (not total)
--led-chain=4	4	Total number of panels in chain
--led-parallel=1	1	Single HUB75 channel (your adapter board)
--led-pixel-mapper="U-mapper"	U-mapper	Creates 2×2 grid from chain [2] [3]
--led-sba=1	1	SERPENTINE BOTTOM ARRANGEMENT (critical!) [4]
--led-slowdown-gpio=4	4	Required for Pi 4 + 4 panels [5] [6]
--led-row-addr-type=3	3	ABC panels (1/32 scan) [6]
--led-multiplexing=0	0	Default for P2 panels (try 0–17 if wrong) [7]
--led-pwm-bits=7	7	Reduces flicker on long chains [3] [6]
--led-brightness=70	70	Indoor use (panels are 700 nits max) [11]

For Text/Scrolling Display (Your ATSB Receiver)

If using displaycurl for scrolling text:

```
cd ~/rpi-rgb-led-matrix  
  
sudo ./displaycurl http://example.com \  
  --led-rows=64 \  
  --led-cols=128 \  
  --led-chain=4 \  
  --led-parallel=1 \  
  --led-pixel-mapper="U-mapper" \  
  --led-slowdown-gpio=4 \  
  --led-row-addr-type=3 \  
  --led-multiplexing=0 \  
  --led-pwm-bits=7 \  
  --led-brightness=70 \  
  --led-sba=1
```

Troubleshooting Guide

If Image Is Wrong/Mirrored:

Symptom	Try This
Image is mirrored/reversed	Remove <code>--led-sba=1</code> flag
Image is scrambled	Try <code>--led-multiplexing=1</code> instead of 0
Image is upside down	Try <code>--led-multiplexing=2</code> or 3
Flickering	Lower <code>--led-pwm-bits</code> to 6 or 5 ^[3]
Color banding	Increase <code>--led-pwm-bits</code> to 8 or 9 ^[3]
Garbage display	Check <code>--led-row-addr-type=3</code> for ABC panels

Test Without Serpentine (if wrong):

```
# Try without --led-sba=1
sudo ./led-image-viewer \
  --led-rows=64 --led-cols=128 --led-chain=4 --led-parallel=1 \
  --led-pixel-mapper="U-mapper" \
  --led-slowdown-gpio=4 --led-row-addr-type=3 \
  --led-multiplexing=0 --led-pwm-bits=7 --led-brightness=70 \
  test.png
```

Test Different Multiplexing Values:

```
# Try multiplexing=0, 1, 2, 3... until correct
--led-multiplexing=0
--led-multiplexing=1
--led-multiplexing=2
```

HUB75 Adapter Board Details

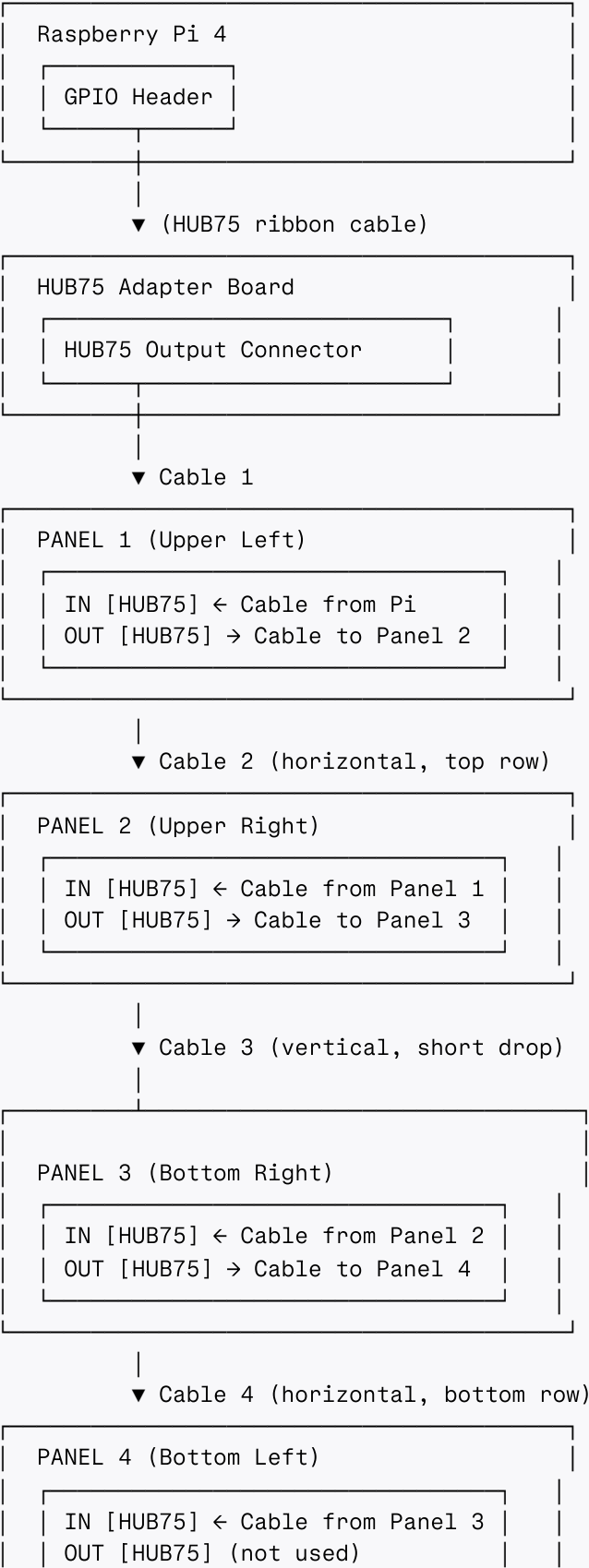
Your Board: AliExpress "Conversion board for Raspberry Pi to HUB75"

- **Type:** Single-channel HUB75 adapter (NOT 3-channel)
- **Connects to:** Raspberry Pi 4 GPIO header (40-pin)
- **Output:** One HUB75 cable to Panel 1 IN
- **Voltage:** 3.3V GPIO → 5V HUB75 (active buffering)
- **Supports:** ABC panels (1/32 scan) ✓

This board requires **U-mapper** (not V-mapper) for 2×2 layout. ^[6] ^[8]

Visual Wiring Diagram (for Mate)

BACK VIEW of Panels (where cables connect):



Cable Summary

Cable #	Length	Connects
Cable 1	10–20 cm	Pi adapter → Panel 1 IN
Cable 2	10–20 cm	Panel 1 OUT → Panel 2 IN (top row)
Cable 3	10–20 cm (short vertical)	Panel 2 OUT → Panel 3 IN (top-right to bottom-right) ✓
Cable 4	10–20 cm	Panel 3 OUT → Panel 4 IN (bottom row, right-to-left)

All cables are standard HUB75 16-pin flat ribbon cables (same as came with panels).

Final Checklist for Mate

- ✓ 4 P2 panels wired in serpentine chain: 1 → 2 → 3 → 4
- ✓ Top row: left-to-right (Panel 1 → Panel 2)
- ✓ Bottom row: right-to-left (Panel 3 → Panel 4)
- ✓ **Must use** `--led-sba=1` for serpentine bottom arrangement
- ✓ **Must use** `--led-pixel-mapper="U-mapper"` (single-channel board)
- ✓ `--led-row-addr-type=3` for ABC panels (1/32 scan)
- ✓ `--led-slowdown-gpio=4` required for Pi 4
- ✓ Panels powered from external 5V supply (not from Pi)

Hope this helps your mate get it working! Let me know if the image is wrong and we can troubleshoot.

*
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1. <https://www.szqaomane.com/store/products/p2-indoor-led-module-256x128mm-p2-led-flexible-panel-led-curved-led-display>
2. <https://ozzmaker.com/getting-started-ozzmaker-16x16-rgb-led-matrix-panel-with-a-raspberry-pi/>
3. <https://www.youtube.com/watch?v=cIMf55Fi-U>
4. <https://forum.arduino.cc/t/ws2811-10-x-10-led-matrix-serpentine-wiring/606106>
5. <https://rpi-rgb-led-matrix.discourse.group/t/panel-p2-5-128x64-hub75-abc-no-display/1124>
6. <https://github.com/hzeller/rpi-rgb-led-matrix/issues/1307>
7. <https://rpi-rgb-led-matrix.discourse.group/t/troubleshooting-a-64x64-hub75-panel-with-a-4-quadrant-split-display/1113>

8. https://marc.merlins.org/perso/arduino/post_2020-01-01_Running-FastLED_-Adafruit_GFX_-and-LEDMatrix-code-on-High-Resolution-RGBPanels-with-a-Raspberry-Pi.html